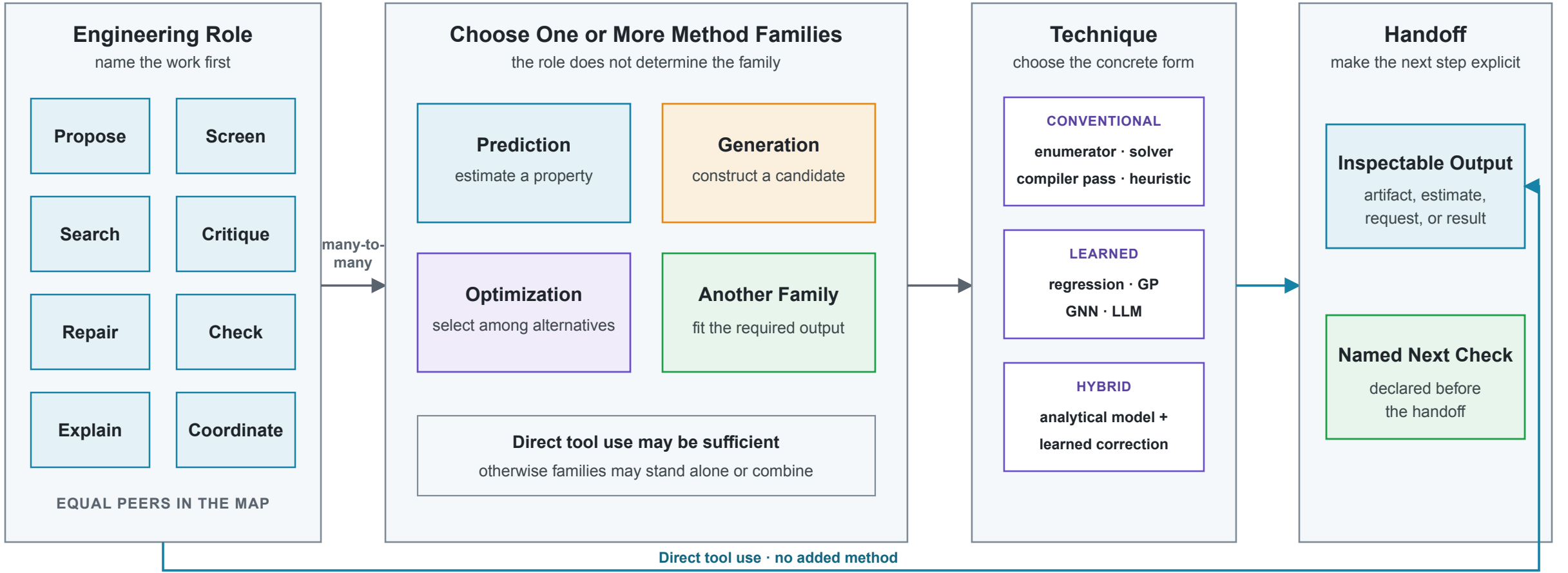

Engineering Role
name the work first
Propose
Screen
Search
Critique
Repair
Check
Explain
Coordinate
EQUAL PEERS IN THE MAP
many-to-many
Choose One or More Method Families
the role does not determine the family
Prediction
estimate a property
Generation
construct a candidate
Optimization
select among alternatives
Another Family
fit the required output
Direct tool use may be sufficient
otherwise families may stand alone or combine
Technique
choose the concrete form
CONVENTIONAL
enumerator · solver
compiler pass · heuristic
LEARNED
regression · GP
GNN · LLM
HYBRID
analytical model +
learned correction
Handoff
make the next step explicit
Inspectable Output
artifact, estimate,
request, or result
Named Next Check
declared before
the handoff
Direct tool use · no added method